ALFIDO MERN STACK INTERNSHIP

TASK 1

Build a RESTful API (Node + Express + MongoDB)

1. Project Overview

This project is a RESTful API designed to manage tasks. It is built using **Node.js** and **Express.js**, with **MongoDB** (via Mongoose) as the database. The API supports full CRUD operations: Creating, Reading, Updating, and Deleting tasks.

- **Repository Link:** https://github.com/sherwin26sanju/task1

2. Technical Stack

- **Backend:** Node.js, Express.js
- **Database:** MongoDB
- **ODM:** Mongoose
- **Testing Tool:** Postman API Client

3. Environment Configuration

Create a .env file in the root directory with the following variables:

```
PORT=3000
MONGODB_URI=mongodb://localhost:27017/taskdb
```

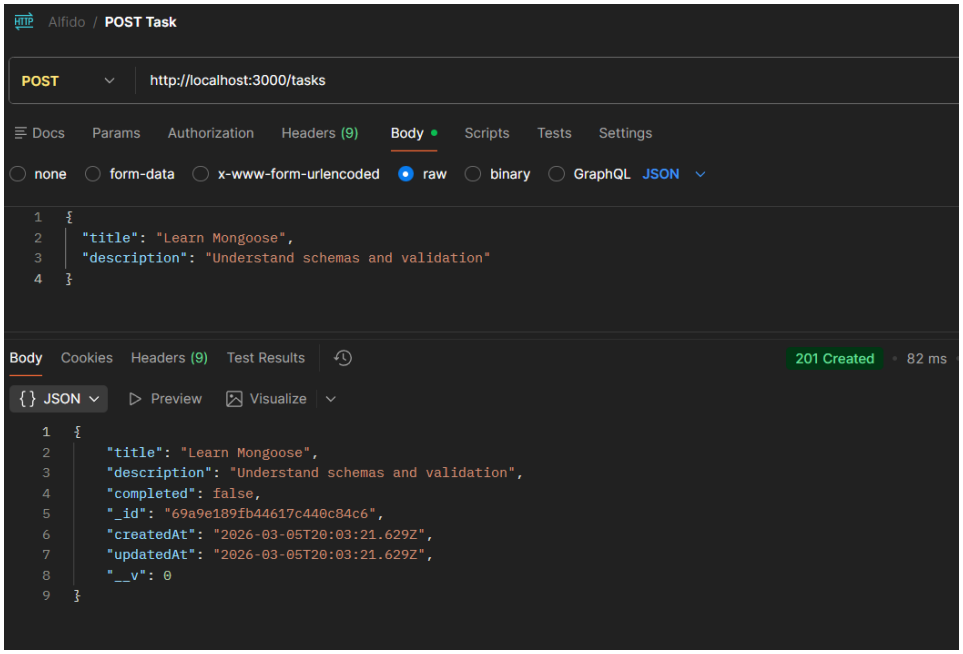
4. API Endpoints & Implementation

A. POST - Create a Task

Used to add a new task to the database with validation for title and description.

- **Endpoint:** POST /tasks

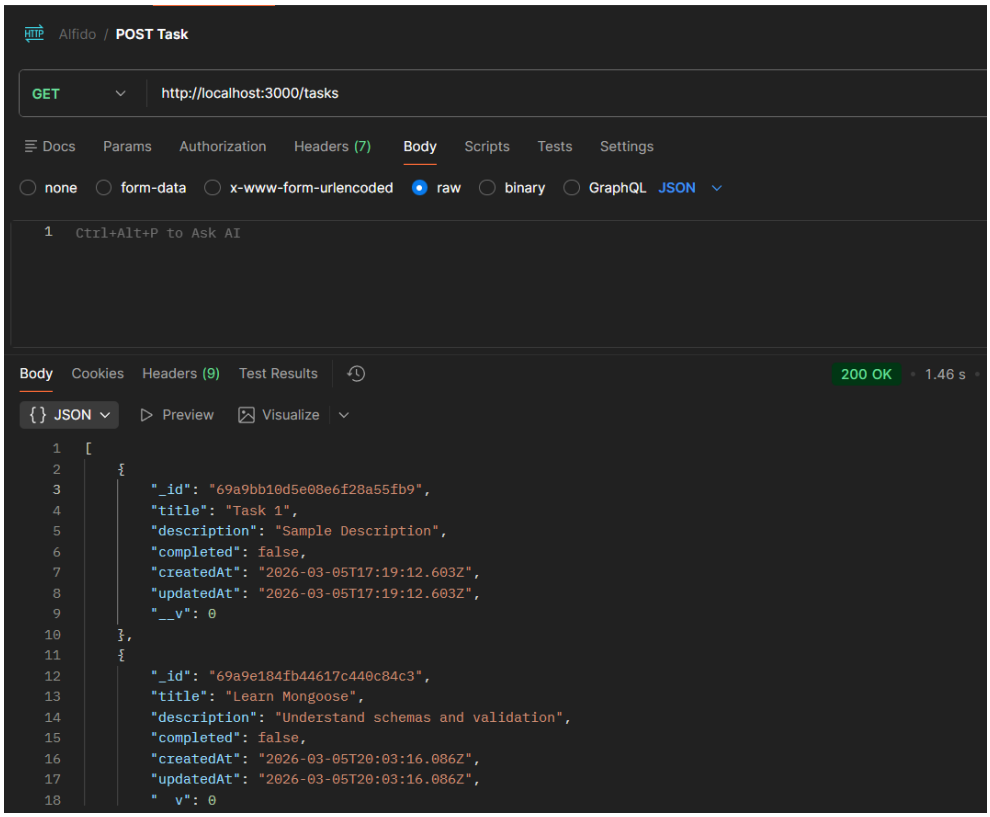
- **Screenshot:**



B. GET - List All Tasks

Fetches the complete list of tasks stored in MongoDB.

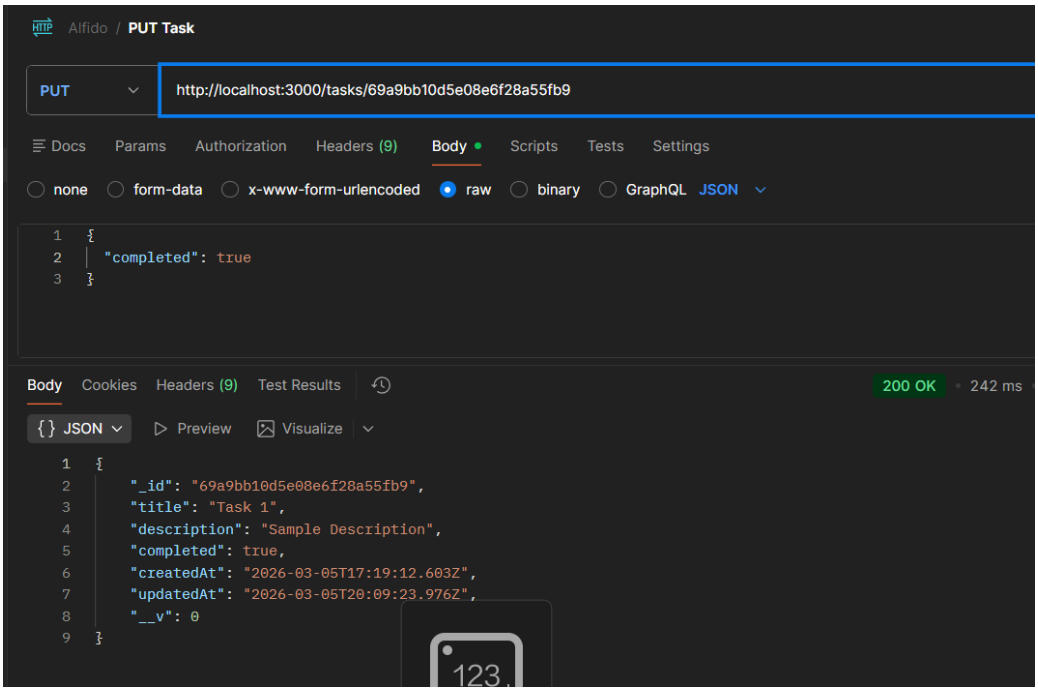
- **Endpoint:** GET /tasks
- **Screenshot:**



C. PUT - Update a Task

Updates the status of a specific task (e.g., marking it as completed: true) using its unique ID.

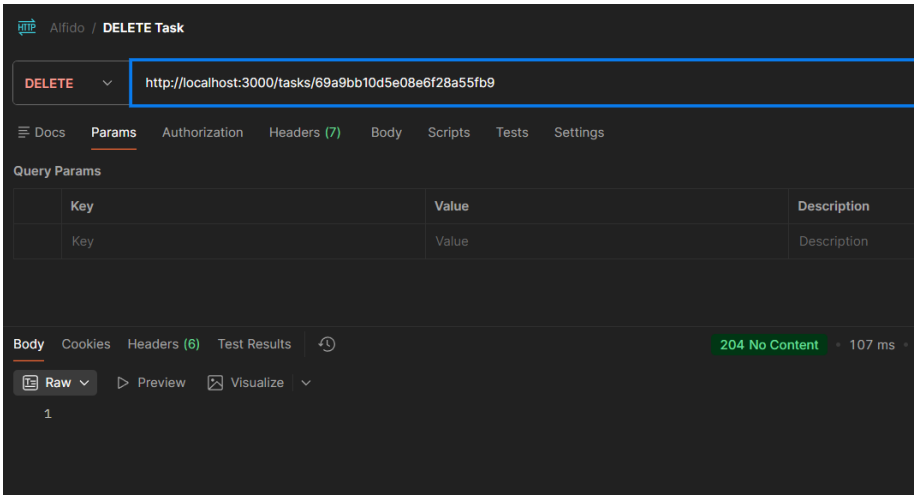
- **Endpoint:** PUT /tasks/:id
- **Screenshot:**



D. DELETE - Remove a Task

Deletes a specific task from the database using its ID.

- **Endpoint:** DELETE /tasks/:id
- **Screenshot:**



5. Mongoose Schema Snippet

The data is structured using the following schema with built-in validation:

```
const taskSchema = new mongoose.Schema(
  {
    title: { type: String, required: [true, "Title is
required"], trim: true, minlength: 3 },
    description: { type: String, maxlength: 500 },
    completed: { type: Boolean, default: false }
  },
  { timestamps: true }
);
```

6. Error Handling & Logging

The API includes:

- Global error-handling middleware to catch 500 Internal Server Errors.
- Validation checks to ensure title and description are provided (returning 400 Bad Request if missing).
- Console logging for server status and database connection success.

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